

Week 1

Quadratic Inequality:

$$x^2 + x - 2 < 0$$

Critical Points: The values of x that make the polynomial equal to zero. These points divide the number line into test intervals where the polynomial is either positive or negative.

PROCEDURE FOR SOLVING POLYNOMIAL INEQUALITIES

- Step 1:** Write the inequality in *standard form*.
- Step 2:** Factor the polynomial.
- Step 3:** Identify critical points.
- Step 4:** Draw a number line with critical points labeled.
- Step 5:** Select a real number in each interval and check sign of product.
- Step 6:** Identify which intervals make the inequality true.
- Step 7:** Write the solution in interval notation.



STUDY TIP

If the original polynomial is < 0 then the interval(s) that yield *negative* numbers should be selected. If the original polynomial is > 0 then the interval(s) that yield *positive* numbers should be selected.

EXAMPLE 1 Solving a Quadratic Inequality

Solve the inequality $x^2 - x > 12$.

Solution:

Write the inequality in standard form.

$$x^2 - x - 12 > 0$$

Factor the left side.

$$(x + 3)(x - 4) > 0$$

Identify the critical points.

$$(x + 3)(x - 4) = 0$$

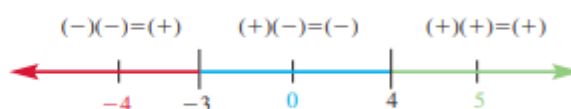
$$x = -3 \text{ and } x = 4$$

Draw a number line with critical points labeled.



Test each interval.

$$(x + 3)(x - 4)$$



Positive intervals make this inequality true.

$$(-\infty, -3) \text{ and } (4, \infty)$$

Write the solution in interval notation.

$$(-\infty, -3) \cup (4, \infty)$$

YOUR TURN Solve the inequality $x^2 - 5x \leq 6$ and express the solution in interval notation.

EXAMPLE 2 Solving a Quadratic Inequality

Solve the inequality $x^2 \leq 4$.

COMMON MISTAKE

Do not take the square root of both sides. You must write the inequality in standard form and factor.



CORRECT

Write the inequality in standard form.

$$x^2 - 4 \leq 0$$

Factor.

$$(x - 2)(x + 2) \leq 0$$

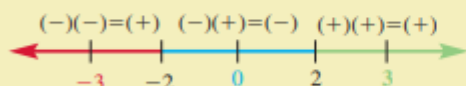
Identify critical points.

$$(x - 2)(x + 2) = 0$$

$$x = 2 \text{ and } x = -2$$

Draw number line and test intervals.

$$(x - 2)(x + 2)$$



Negative intervals make the inequality true.

$$(-2, 2)$$

The endpoints, $x = -2$ and $x = 2$, satisfy the inequality so they are included in the solution and

expressed with brackets. $[-2, 2]$



INCORRECT

Write the inequality.

$$x^2 \leq 4$$

Take the square root of both sides.

$$x \leq 2 \quad \text{ERROR}$$

INCORRECT solution:

$$(-\infty, 2]$$

When solving quadratic inequalities, you must first write in standard form and then factor to identify critical points.

EXAMPLE 3 Solving a Quadratic Inequality

Solve the inequality $x^2 > -5x$.

COMMON MISTAKE

A common mistake is to divide by x because if x is negative the sign must be reversed. Always start by writing the inequality in standard form and factoring to determine critical points.



CORRECT

Write inequality in standard form.

$$x^2 + 5x > 0$$

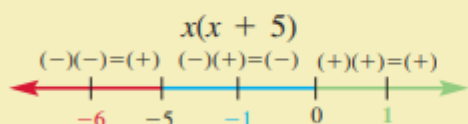
Factor.

$$x(x + 5) > 0$$

Identify critical points.

$$x = 0, x = -5$$

Draw number line and test intervals.



Positive intervals satisfy the inequality.

$$(-\infty, -5) \text{ and } (0, \infty)$$

Express the solution in interval notation.

$$(-\infty, -5) \cup (0, \infty)$$



INCORRECT

Write the original inequality.

$$x^2 > -5x$$

Divide both sides by x .

$$x > -5 \quad \text{ERROR}$$

INCORRECT solution:

$$(-5, \infty)$$

Dividing by x is the mistake. If x is negative, the inequality sign must be reversed. And what if x is zero?

The solution to the inequality $x^2 > -5x$ is $(-\infty, -5) \cup (0, \infty)$.

Rational Inequality:

$$\frac{x-3}{x^2-4} \geq 0$$

In rational inequalities, the critical points are any values that make either the numerator or denominator equal to zero.



EXAMPLE 4 Solving a Rational Inequality

Solve the inequality $\frac{x-3}{x^2-4} \geq 0$.

Solution:

Factor denominator.

$$\frac{(x-3)}{(x-2)(x+2)} \geq 0$$

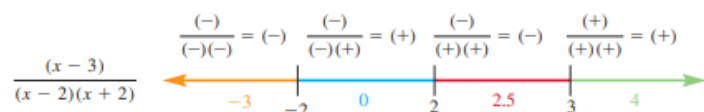
Identify critical points.

$$x = -2, x = 2, x = 3$$

Draw a number line and label critical points.



Test intervals.



Positive intervals satisfy this inequality.

$$(-2, 2) \text{ and } (3, \infty)$$

Since this inequality is greater than or equal to, we include $x = 3$ in our solution because it satisfies the inequality. However, the points $x = -2$ and $x = 2$ are not included in the solution because they make the denominator zero.

The solution is $(-2, 2) \cup [3, \infty)$.

YOUR TURN Solve the inequality $\frac{x+2}{x-1} \leq 0$.

Answer: $[-2, 1)$

EXAMPLE 5 Solving a Rational Inequality

Solve the inequality $\frac{x}{x+2} < 3$.

COMMON MISTAKE

Do not cross multiply.



CORRECT

Subtract 3 from both sides.

$$\frac{x}{x+2} - 3 < 0$$



INCORRECT

Do not cross multiply.

$$x < 3(x+2) \quad \text{ERROR}$$

Write as a single rational function.

$$\frac{x - 3(x+2)}{x+2} < 0$$

Eliminate parentheses.

$$\frac{x - 3x - 6}{x+2} < 0$$

Simplify numerator.

$$\frac{-2x - 6}{x+2} < 0$$

Factor numerator.

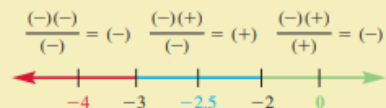
$$\frac{-2(x+3)}{x+2} < 0$$

Identify the critical points.

$$x = -3 \text{ and } x = -2$$

Draw number line and test intervals.

$$\frac{-2(x+3)}{x+2} < 0$$



Negative intervals satisfy the inequality. $(-\infty, -3)$ and $(-2, \infty)$.

Note that $x = -2$ is not included in the solution because it makes the denominator zero and $x = -3$ is not included because it does not satisfy the strict inequality.

The solution is:

$$(-\infty, -3) \cup (-2, \infty)$$

Eliminate parentheses.

$$x < 3x + 6$$

Collect terms.

$$-2x < 6$$

Divide by -2 , and reverse sign.

$$x > -3 \quad \text{INCORRECT}$$

INCORRECT solution:

$$(-3, \infty)$$

This solution is incorrect because the sign of $(x+2)$ can be positive or negative depending on the value of x . It is not appropriate to multiply both sides of an inequality by a variable expression.

Practice Questions

In Exercises 1–22, solve the polynomial inequality and express the solution set in interval notation.

1. $x^2 - 3x - 10 \geq 0$
2. $x^2 + 2x - 3 < 0$
3. $p^2 + 4p < -3$
4. $p^2 - 2p \geq 15$
5. $2t^2 - 3 \leq t$
6. $3t^2 \geq -5t + 2$
7. $5v - 1 > 6v^2$
8. $12t^2 < 37t + 10$
9. $2s^2 - 5s \geq 3$
10. $8s + 12 \leq -s^2$
11. $y^2 + 2y \geq 4$
12. $y^2 + 3y \leq 1$
13. $u^2 \geq 3u$
14. $u^2 \leq -4u$
15. $-2x \leq -x^2$
16. $-3x \leq x^2$
17. $x^2 > 9$
18. $x^2 \geq 16$
19. $z^2 > -16$
20. $z^2 \geq -2$
21. $y^2 < -4$
22. $y^2 \leq -25$

In Exercises 23–40, solve the rational inequality and graph the solution on the real number line.

23. $-\frac{3}{x} \leq 0$
24. $\frac{3}{x} \leq 0$
25. $\frac{y}{y+3} > 0$
26. $\frac{y}{2-y} \leq 0$
27. $\frac{s+1}{4-s^2} \geq 0$
28. $\frac{s+5}{4-s^2} \leq 0$
29. $\frac{2u^2+u}{3} < 1$
30. $\frac{u^2-3u}{3} \geq 6$
31. $\frac{3t^2}{t+2} \geq 5t$
32. $\frac{-2t-t^2}{4-t} \geq t$
33. $\frac{3p-2p^2}{4-p^2} < \frac{3+p}{2-p}$
34. $-\frac{7p}{p^2-100} \leq \frac{p+2}{p+10}$
35. $\frac{x^2}{5+x^2} < 0$
36. $\frac{x^2}{5+x^2} \leq 0$
37. $\frac{x^2+10}{x^2+16} > 0$
38. $-\frac{x^2+2}{x^2+4} < 0$
39. $\frac{v^2-9}{v-3} \geq 0$
40. $\frac{v^2-1}{v+1} \leq 0$

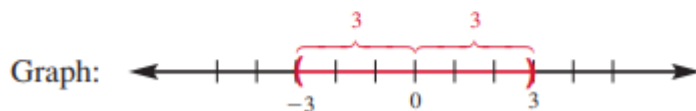
Absolute Inequality:

$$|x| < 3$$

If we interpret this inequality as distance, we ask what numbers are less than 3 units from the origin? We can represent the solution in the following ways.

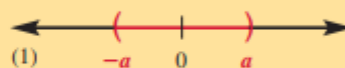
Inequality notation: $-3 < x < 3$

Interval notation: $(-3, 3)$

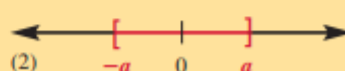


PROPERTIES OF ABSOLUTE VALUE INEQUALITIES

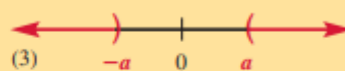
1. $|x| < a$ is equivalent to $-a < x < a$



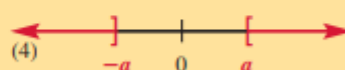
2. $|x| \leq a$ is equivalent to $-a \leq x \leq a$



3. $|x| > a$ is equivalent to $x < -a$ or $x > a$



4. $|x| \geq a$ is equivalent to $x \leq -a$ or $x \geq a$



EXAMPLE 6 Solving an Inequality Involving an Absolute ValueSolve the inequality $|3x - 2| \leq 7$.**Solution:**We apply property (2) and squeeze the absolute value expression between -7 and 7 .

$$-7 \leq 3x - 2 \leq 7$$

Add 2 to all three parts.

$$-5 \leq 3x \leq 9$$

Divide all three parts by 3.

$$-\frac{5}{3} \leq x \leq 3$$

The solution in interval notation is $\left[-\frac{5}{3}, 3\right]$.

Graph:

**YOUR TURN** Solve the inequality $|2x + 1| < 11$.**Answer:** Inequality notation: $-6 < x < 5$ Interval notation: $(-6, 5)$

It is often helpful to note that for absolute value inequalities

- $\blacksquare$ *less than* inequalities can be written as a single statement (see Example 6).
- $\blacksquare$ *greater than* inequalities must be written as two statements (see Example 7).

**EXAMPLE 7** Solving an Inequality Involving an Absolute ValueSolve the inequality $|1 - 2x| > 5$.**Solution:**

Apply property (3).

$$1 - 2x < -5 \quad \text{or} \quad 1 - 2x > 5$$

Subtract 1 from all expressions.

$$-2x < -6 \qquad -2x > 4$$

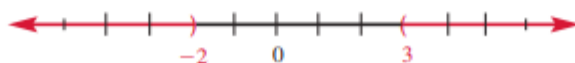
Divide by -2 and reverse the sign.

$$x > 3 \qquad x < -2$$

Express the solution in interval notation.

$$(-\infty, -2) \cup (3, \infty)$$

Graph:

**YOUR TURN** Solve the inequality $|5 - 2x| \geq 1$.**Answer:** Inequality notation: $x \leq 2$ or $x \geq 3$ Interval notation: $(-\infty, 2] \cup [3, \infty)$

Practice Questions

In Exercises 27–46, solve the inequality and express the solution in interval notation and graph.

27. $|x| < 7$

28. $|y| < 9$

29. $|y| \geq 5$

30. $|x| \geq 2$

31. $|x + 3| < 7$

32. $|x + 2| \leq 4$

33. $|x - 4| > 2$

34. $|x - 1| < 3$

35. $|4 - x| \leq 1$

36. $|1 - y| < 3$

37. $|2x| > -3$

38. $|2x| < -3$

39. $|4 - 3x| \geq 0$

40. $|4 - 3x| \geq 1$

41. $|1 - 2x| < \frac{1}{2}$

42. $\left| \frac{2 - 3x}{5} \right| \geq \frac{2}{5}$

43. $|2.6x + 5.4| < 1.8$

44. $|3.7 - 5.5x| > 4.3$

45. $|x^2 - 1| \leq 8$

46. $|x^2 + 4| \geq 29$